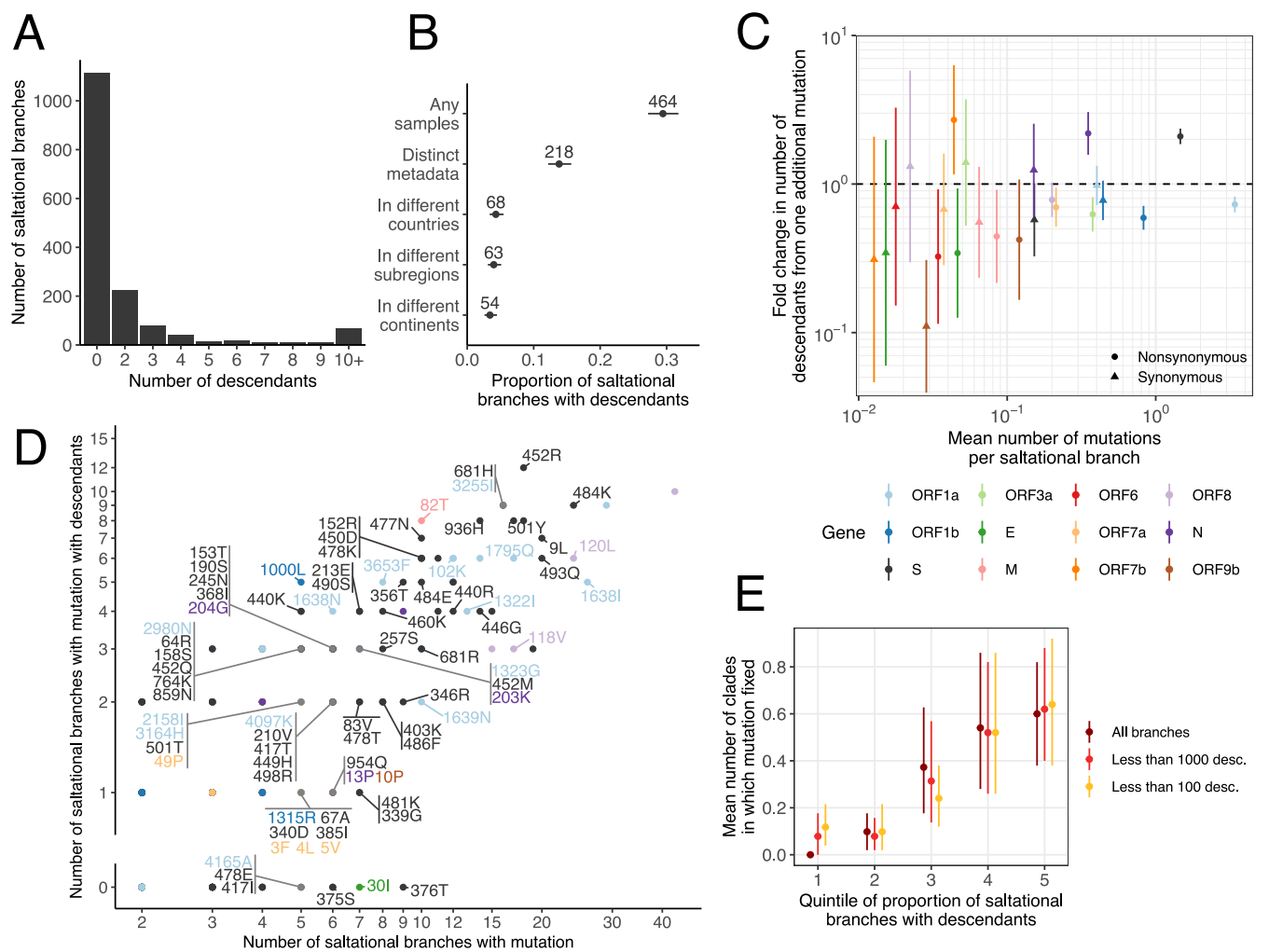

A
Number of saltational branches
1000
800
600
400
200
0
0 2 3 4 5 6 7 8 9 10+
Number of descendants
B
Any samples
Distinct metadata
In different countries
In different subregions
In different continents
464
218
68
63
54
0.0 0.1 0.2 0.3
Proportion of saltational branches with descendants
C
Fold change in number of descendants from one additional mutation
10^1
10^0
10^{-1}
10^{-2}
10^{-1} 10^0
Mean number of mutations per saltational branch
Nonsynonymous
Synonymous
Gene
ORF1a ORF3a ORF6 ORF8
ORF1b E ORF7a N
S M ORF7b ORF9b
D
Number of saltational branches with mutation with descendants
15 12 10 9 8 7 6 5 4 3 2 1 0
452R
681H
3255I
484K
82T
152R
450D
478K
477N
936H
501Y
9L
120L
1795Q
153T
190S
245N
368I
204G
1000L
213E
490S
3653F
102K
493Q
1638I
440K
1638N
356T
484E
440R
1322I
460K
446G
118V
2980N
64R
158S
452Q
764K
859N
257S
681R
1323G
452M
203K
346R
1639N
2158I
3164H
501T
49P
4097K
210V
417T
449H
498R
83V
478T
403K
486F
954Q
13P10P
481K
339G
1315R 67A
340D 385I
3F 4L 5V
4165A
478E
417I
375S
30I
376T
2 3 4 5 6 7 8 9 10 12 15 20 30 40
Number of saltational branches with mutation
E
Mean number of clades in which mutation fixed
0.8
0.6
0.4
0.2
0.0
1 2 3 4 5
Quintile of proportion of saltational branches with descendants
All branches
Less than 1000 desc.
Less than 100 desc.